



TECHNICAL SKILLS

Tools	Python, R, SQL, PostgreSQL, MySQL, Azure, Git, MariaDB
File Formats	XML, YAML, Markdown, JSON
Deploy env	HuggigFace, Railway, Vercel, Render
Operating systems	Linux, Windows
Frameworks/Platforms	LangChain, FastAPI, Keras, Docker, Streamlit, Shiny, VS Code, Jupyter
Modeling methods	Linear Regression, Logistic Regression, Ridge/Lasso Regression, Random Forests, XGBoost, kNN, SVM, PCA, Neural Networks (CNN)/(RNN)/(LSTM)/(VAE), Time Series Models, Seaborn, XGBoost, LightGBM
Familiar with	Apache Superset, HuggingFace, Prophet forecasting, Supabase, MongoDB, BERT, PySpark, C++, HTML, Java, Modular programming, AWS, Azure
Vector database	Milvus, Weaviate, FAISS, ChromaDB

EDUCATION

M.Sc. in Data Science, Université du Luxembourg
M.Sc. in Physics, Sharif University of Technology, Tehran
B.Sc. in Physics, Semnan University, Semnan

PROJETS

GAME FEATURES DEVELOPMENT

Designed and implemented matchmaking features to improve player engagement by intelligently pairing users. The system integrates Bot level analysis and latency measurement to ensure balanced and responsive game-play.

PM2 Python SQL tmux OOp Visual Studio LaTeX

AI AGENT ASSISTANT

Developed a smart agent system capable of answering questions based on content extracted from user-provided documents such as PDF, TXT, or Word files. The system leverages Retrieval-Augmented Generation (RAG) and a vector database to deliver accurate, context-aware responses. Full-stack development including frontend and backend logics. In the next phase, user-specific functionality can be added, including personalized search history for each unique user, powered by MinIO S3 buckets.

FastAPI Python SQL OpenAI API Chroma DB minio

VOICE AI ASSISTANT

Built a real-time voice assistant capable of handling inbound calls and responding to user queries using natural language. Leveraged LLMs in combination with Telnyx and LiveKit Cloud for robust voice-to-text and text-to-speech functionality. Designed for easy conversation and knowledge retrieval. System supports multi-language identification.

LiveKit Cloud Telnyx FastAPI Python OpenAI API RAG

REAL ESTATE AI ASSISTANT

Developed API endpoints in FastAPI framework for a real estate platform that assists realtors in identifying suitable property listings for clients. Integrated a chatbot for intelligent Q&A and database search to streamline record recovery.

FastAPI Python JWT S3 buckets PDF/Excel/JPG Parsing openstreetmap

VARIATIONAL AUTOENCODERS FOR CHAOTIC SYSTEMS

Implemented a Variational Autoencoder (VAE) to learn and generate trajectories of chaotic dynamical Henon-Heiles Hamiltonian system. The model captures latent structure from test/train data (generated by molecular dynamics) and reconstructs realistic patterns based on feature space. Emphasis is on improvement through regularization and validation techniques.

VAE Neural Networks Regularization Train/Test/Validation Split

BIG DATA ANALYTICS

Transforming Big-data analytics scripts from Scala to PySpark

AWS PySpark Big Data